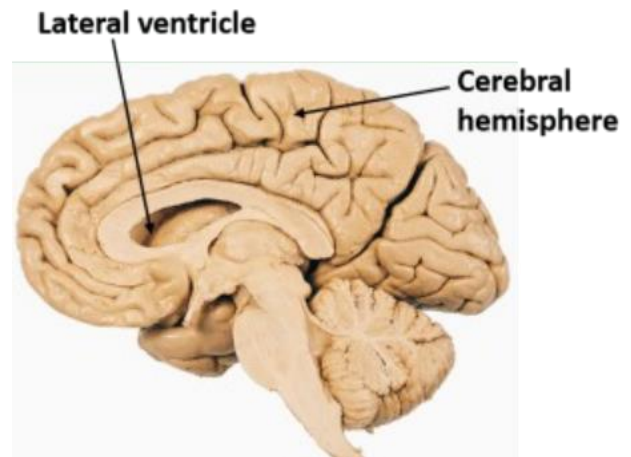


Lecture (9): Meninges and ventricular system II

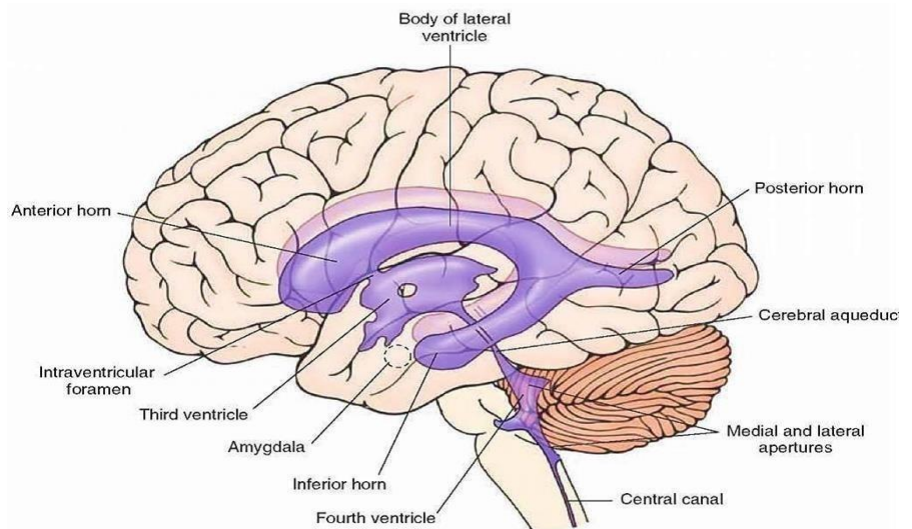
Ventricular System:

- The **ventricles** are **four** fluid-filled cavities lined with **ependyma**, are located within the brain; these are **two** lateral ventricles, **3rd** ventricle and **4th** ventricle.
- The **two lateral ventricles** communicate through the **interventricular foramina of Monro** with the **3rd** ventricle.
- The **3rd** ventricle is connected to the **4th** ventricle by **cerebral aqueduct (aqueduct of Sylvius)** in **midbrain**.
- The **4th** ventricle is continuous below with:
 - **Central canal** of the spinal cord.
 - **Subarachnoid space** through **three foramina**.
- The central canal in the spinal cord has a small dilatation at its inferior end, referred to **terminal ventricle**.



Lateral Ventricle:

- There are two large lateral ventricles, one is present in **each cerebral hemisphere**.
- The ventricle is **C-shaped** cavity.
- The lateral ventricle communicates with the cavity of the **3rd** ventricle through the **interventricular foramen of Monro**.
- **Choroid plexuses** of this ventricle are derived from anterior & posterior choroidal arteries.



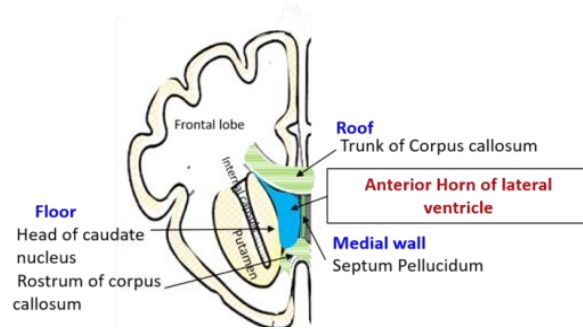
- Parts:

- The **anterior horn:**

- Extends forward into the frontal lobe.
- It is surrounded superiorly, inferiorly and anteriorly by parts of corpus callosum.
- Lateral wall by **head** of caudate.
- Medial wall by **septum pellucidum**.

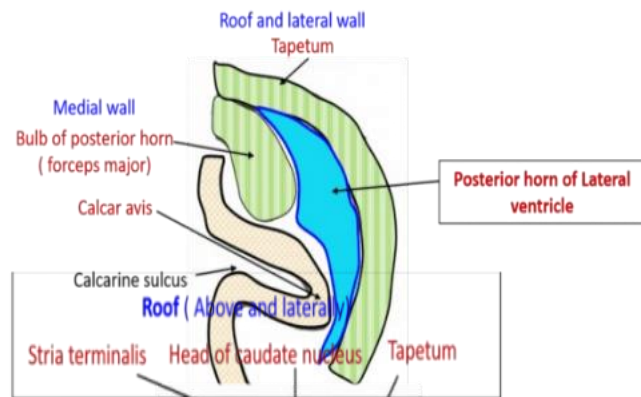
- The body (central part):

- It occupies the **parietal lobe**.
- Its **roof** is formed by the body of corpus callosum.
- Its **floor** is formed by the body of caudate nucleus, thalamus, body of fornix.
- Medial wall by septum pellucidum.



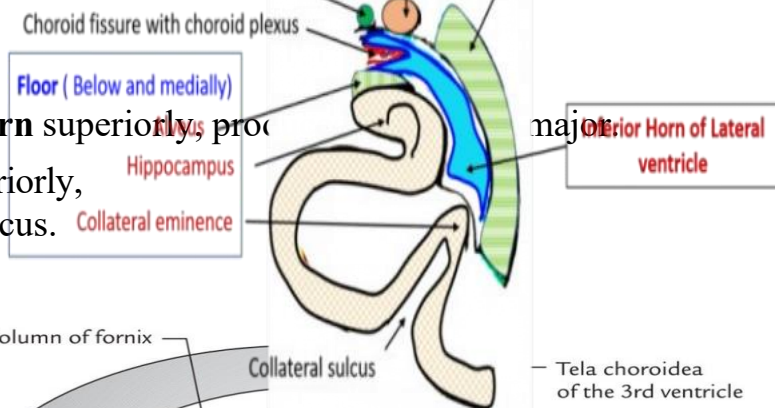
- The **posterior horn:**

- Extends posteriorly into the occipital lobe.
- Its roof & lateral wall are formed by **tapetum** of corpus callosum.



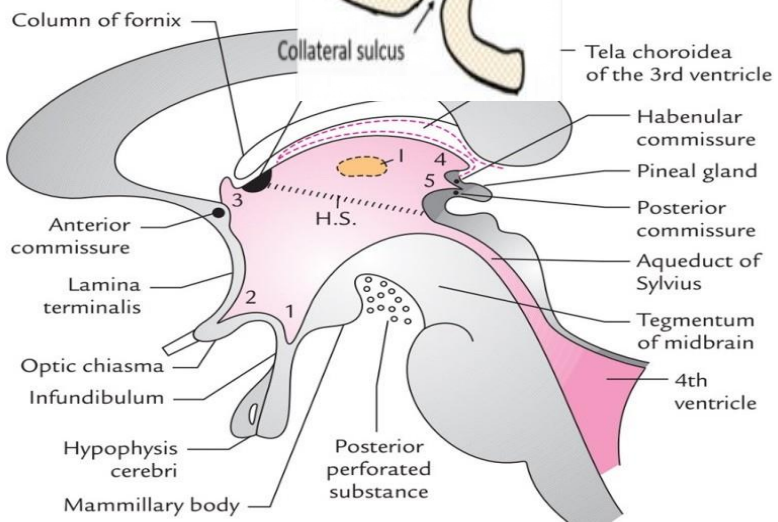
- Its medial wall is formed into **two elevations:**

1. The **bulb** of **posterior horn** superiorly, produced by calcarine sulcus.
2. The **calcar Avis** inferiorly, produced by calcarine sulcus.



- The **inferior horn:**

- Extends anteriorly into the **temporal lobe**.
- Its **roof** is formed by: **Tail of caudate & stria terminalis** of amygdaloid nucleus.
- Its **floor** is formed by



Hippocampus & Collateral eminence.

Third Ventricle:

- The 3rd is the **cavity** of diencephalon.
- It is a midline slit-like cleft between the two thalami.
- **Choroid plexuses** of this ventricle are derived from anterior & posterior choroidal arteries.
- **Communications:**
 - With the **lateral ventricles** through **interventricular foramina of Monro**.
 - With the 4th ventricle through **cerebral aqueduct of Sylvius**.
- **Boundaries:**
 - **Roof:** Created by the **ependyma & Tela choroidea** that stretch across the upper limits of 2 thalami.
 - **Floor:** Created from before backwards by: **Optic chiasma, infundibulum, tuber cinereum, mammillary bodies and tegmentum of midbrain**.
 - **Lateral wall:** thalamus, hypothalamic sulcus & hypothalamus.
 - **Anterior:** lamina terminalis, Anterior commissure & ant. column of fornix.
 - **Posterior:** Pineal gland & Posterior commissure.
- The cavity of the 3rd ventricle extends into the surrounding structures as pocket-like protrusions referred to as **recesses**. All these are as follows: **Optic (or chiasmatic), Infundibular, Suprapineal and Pineal recesses**

Fourth ventricle:

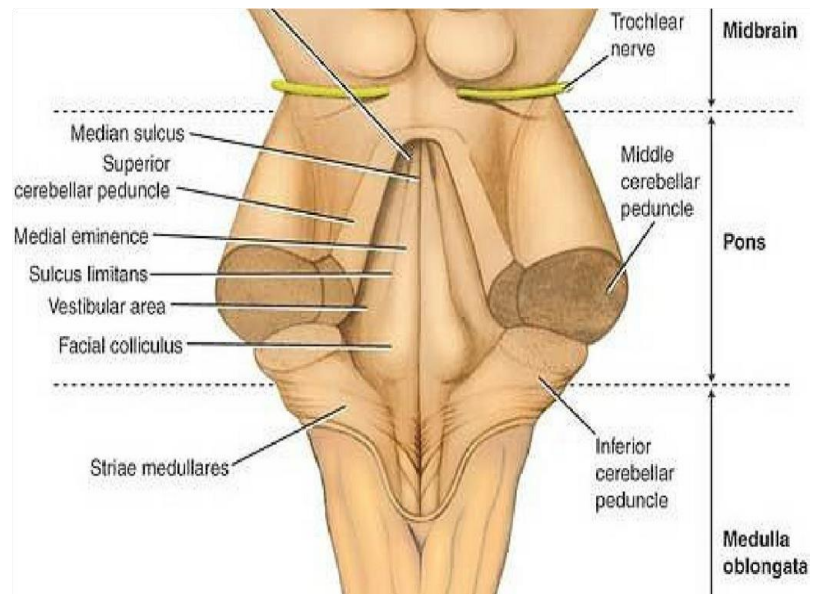
- It is the cavity of hindbrain.
- The 4th ventricle is a **rhomboid-shaped fossa**.
- It is situated in front of cerebellum and behind the pons and the upper half of the medulla oblongata.
- Choroid plexus of this ventricle is derived from posterior inferior cerebellar artery.
- **Communications:**
 - It is continuous above with the 3rd ventricle through the cerebral aqueduct of

Sylvius.

- Below with the central canal of the medulla oblongata and the spinal cord.
- **Foramina of fourth ventricle:** It communicates with subarachnoid space through a **single median foramen of Magendie** and **two lateral foramina of Luschka**.

▪ Boundaries:

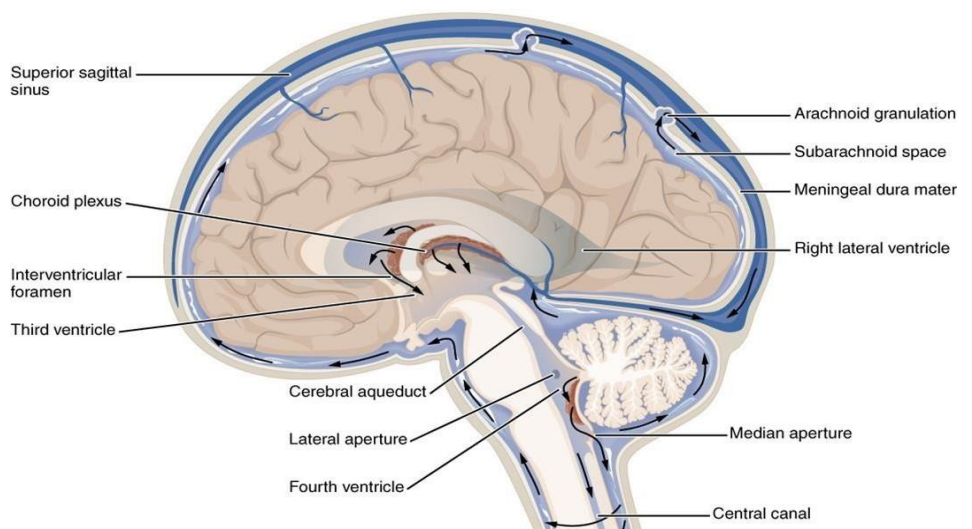
- The **superolateral** borders are formed by Rt & Lt superior cerebellar peduncles.
- The **inferolateral** borders are formed by Rt & Lt inferior cerebellar peduncles
- **Roof:** superior & inferior medullary velum separating it from cerebellum.
- **Floor (Rhomboid fossa):** is a diamond-shaped depression on the dorsal surface of pons and medulla, is divided by stria medullaris into upper pontine part & lower medullary part. Floor is divided into symmetrical Rt & Lt halves by the posterior median sulcus.



Cerebrospinal Fluid (CSF):

- The cerebrospinal fluid is found in the **ventricles** of the brain and in the **subarachnoid space** around the brain and spinal cord.
- It has a volume of about 135 ml. It is a clear, colorless fluid.

- **Functions:**
 - Protects the central nervous system from trauma.
 - Provides mechanical buoyancy and support for the brain.
- Serves as a reservoir and assists in the regulation of the



contents of the skull.

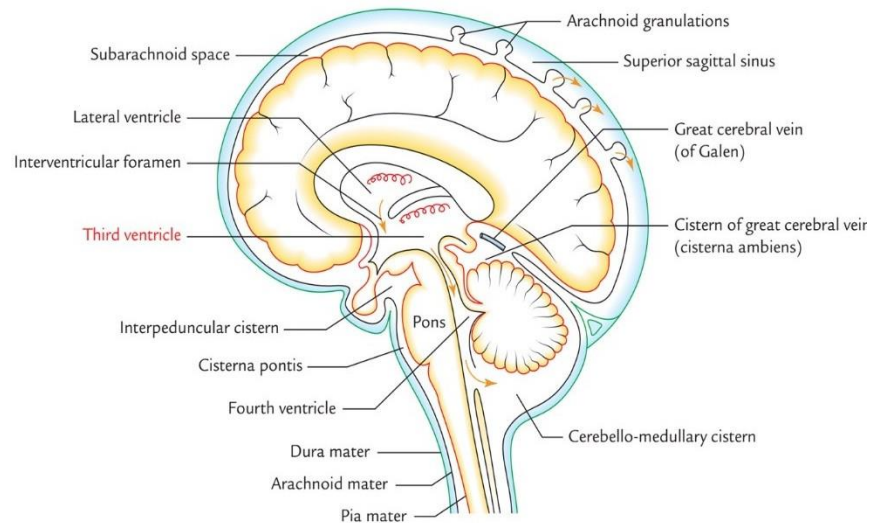
- Nourishes the central nervous system.
- Removes metabolites from the central nervous system.

➤ Formation:

- CSF is produced mainly by the **choroid plexuses** of the four ventricles of the brain and from the brain substance through the perivascular spaces.
- Choroid plexus is a mass of blood capillaries invaginating the pia mater and ependyma.

➤ Circulation:

- CSF is secreted from the choroid plexuses in the ventricles.
- The fluid passes from the lateral ventricles into the **3rd ventricle** through the interventricular foramina.
- It then passes into the **4th ventricle** through the cerebral aqueduct of midbrain.
- From the **4th ventricle**, the fluid passes slowly through the median aperture of Magendie and the lateral foramina of Luschka and enters the subarachnoid space.
- The fluid then moves in subarachnoid around the brain and the spinal cord.



➤ Absorption:

- The main sites for the absorption of the cerebrospinal fluid are the **arachnoid villi** that projects into the **superior sagittal sinus**.
- The arachnoid villi tend to be grouped together to form **arachnoid granulations**.

Blood-Brain Barrier (BBB):

- BBB is a highly selective permeable membrane between the blood vessels & the brain. Structure of BBB: Endothelial cells of in the capillary wall which are connected by tight junctions, Continuous basement membrane and Foot processes of the astrocytes.